

3 - Secure Mail Server Setup and Email Inbox Configuration

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Course: Unix-based Internet Applications

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Abstract

During the lab, a full email server with proper environment was configured, deployed, secured and tested, starring modern real-world tools and practices.

The setup of main tools is listed:

- Postfix - a mail server that handles SMTP.
- Dovecot - for IMAP/POP3 email retrieval.
- Roundcube - for the email interface (a local hosted web-app).
- Virtual mails with password-based file structure were created to allow single cross-functional access between tools.

The setup of security technologies:

- OpenDKIM (DomainKey Identified Mail) - for signing mail messages with a domain key.
- SPF (Sender Policy Framework) - for authenticating sending server to a starred domain.
- TLS/SSL - for security across different processes to ensure that email is transmitted, authenticated across encrypted channels.
- Firewall rules were updated to open only necessary ports (restrict access to the chosen ones).

By the end of the lab, email pipeline—sending, receiving, storing, retrieving, and logging messages—was functioning securely and reliably.

Overview of Email Server Architecture

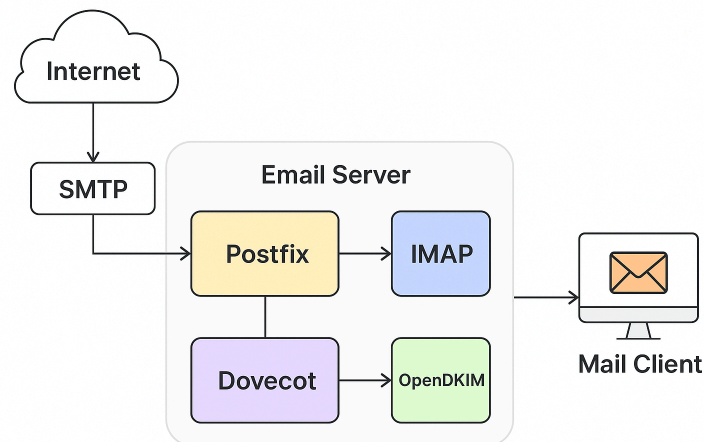


Figure 1. Architecture [GPT was used to create such a diagram]

Brief Description of Protocols (SMTP, IMAP, POP3)

SMTP (Single Mail Transfer Protocol) is used only to transfer (no storage, no downloading) messages from client to server e.g., from the Innopolis.University email server to GMAIL server. Used on the web. Placed on port 25. Uses TLS/SSL for secure transmission: 465 port SSL / port 587 TLS.

POP3 (Post Office Protocol) - downloads email messages from the server to local device, making them available offline, and deletes them from the server. Placed on port 110. Less secure than IMAP, but still TLS/SSL ports (995)

IMAP (Internet Mail Access Protocol) - access, manage of email messages on a remote server and provides synchronization between devices. Places on port 143. Requires internet connection. Security is configured on port 993.

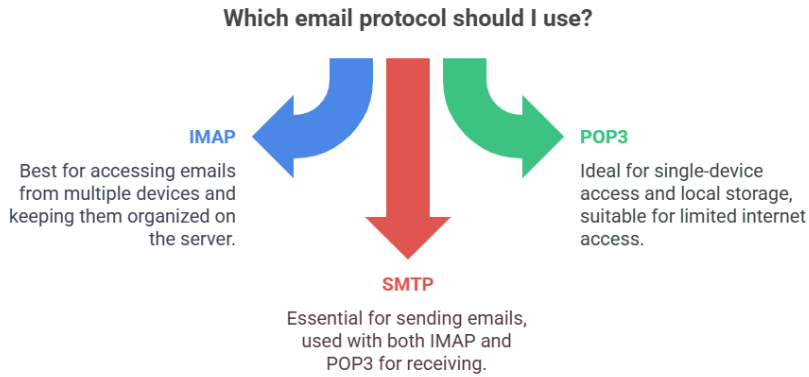


Figure 2. [<https://www.geeksforgeeks.org/computer-networks/imap-vs-pop3-vs-smtp/>]

Input

Hostname: email

Domain Used: mail.example.com

Server IP: 192.168.1.100

Part 1: System Preparation and Initial Setup

Task 1.1: Update System Packages

- Updated the system to ensure all packages are current:

```
sudo apt update
sudo apt upgrade -y
```

Task 1.2: Configure Hostname

- Set system hostname (replace mail.example.com with template domain):

```
sudo hostnamectl set-hostname mail.example.com
```

- Edited `/etc/hosts` to include my hostname:

```
sudo nano /etc/hosts
```

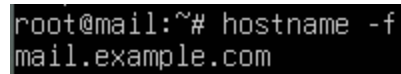
- Added the following line:

```
192.168.1.100 mail.example.com mail
```

Task 1.4: Install Essential Tools

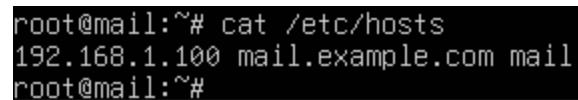
```
sudo apt install -y curl wget git net-tools dnsutils telnet
```

- Deliverable 1.1: Take a screenshot showing the output of `hostname -f` and `cat /etc/hosts`.



```
root@mail:~# hostname -f
mail.example.com
```

Figure 3.



```
root@mail:~# cat /etc/hosts
192.168.1.100 mail.example.com mail
root@mail:~#
```

Figure 4.

Part 2: Postfix SMTP Server Installation and Configuration

Task 2.1: Install Postfix

- Installed Postfix mail server:

```
sudo apt install -y postfix
```

- During installation:
 - Selected "Internet Site" as the configuration type
 - Entered my domain name (example.com)

Task 2.2: Configure Postfix Main Configuration

- Edited the main Postfix configuration file:

```
sudo nano /etc/postfix/main.cf
```

- Modified the following parameters:

```
# Basic Settings
myhostname = mail.example.com
mydomain = example.com
myorigin = $mydomain
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain

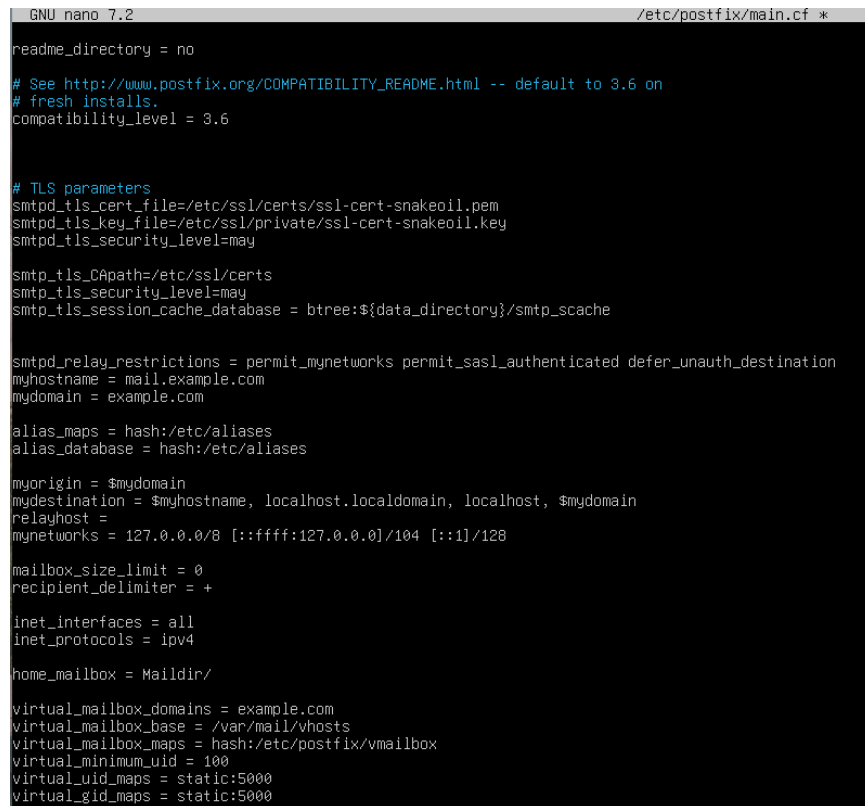
# Network Settings
inet_interfaces = all
inet_protocols = ipv4

# Mail Storage
home_mailbox = Maildir/
mailbox_size_limit = 0
recipient_delimiter = +

# Virtual Mailbox Settings (for virtual users)
virtual_mailbox_domains = example.com
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
virtual_gid_maps = static:5000
```

SMTP Client Restrictions

```
smtpd_relay_restrictions = permit_mynetworks,  
    permit_sasl_authenticated,  
    defer_unauth_destination
```



```
GNU nano 7.2 /etc/postfix/main.cf *  
  
readme_directory = no  
  
# See http://www.postfix.org/COMPATIBILITY_README.html -- default to 3.6 on  
# fresh installs.  
compatibility_level = 3.6  
  
# TLS parameters  
smtpd_tls_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem  
smtpd_tls_key_file=/etc/ssl/private/ssl-cert-snakeoil.key  
smtpd_tls_security_level=may  
  
smtp_tls_CApath=/etc/ssl/certs  
smtp_tls_security_level=may  
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache  
  
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination  
myhostname = mail.example.com  
mydomain = example.com  
  
alias_maps = hash:/etc/aliases  
alias_database = hash:/etc/aliases  
  
myorigin = $mydomain  
mydestination = $myhostname, localhost.localdomain, localhost, $mydomain  
relayhost =  
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128  
  
mailbox_size_limit = 0  
recipient_delimiter = +  
  
inet_interfaces = all  
inet_protocols = ipv4  
  
home_mailbox = Maildir/  
  
virtual_mailbox_domains = example.com  
virtual_mailbox_base = /var/mail/vhosts  
virtual_mailbox_maps = hash:/etc/postfix/vmailbox  
virtual_minimum_uid = 100  
virtual_uid_maps = static:5000  
virtual_gid_maps = static:5000
```

Figure 5.

Task 2.3: Configure SMTP Authentication

- Added SASL authentication settings to `/etc/postfix/main.cf`:

SMTP Authentication

```
smtpd_sasl_auth_enable = yes  
smtpd_sasl_type = dovecot  
smtpd_sasl_path = private/auth  
smtpd_sasl_security_options = noanonymous
```

```
smtpd_sasl_local_domain = $myhostname
broken_sasl_auth_clients = yes
```

```
home_mailbox = Maildir/

virtual_mailbox_domains = example.com
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
virtual_gid_maps = static:5000

smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_auth_clients = yes
```

Figure 6.

Task 2.4: Configure Master Process

- Edited `/etc/postfix/master.cf` to enable submission port:

```
sudo nano /etc/postfix/master.cf
```

- Uncommented and configured the submission service (port 587):

```
submission inet n      -    y    -    -    smtpd
-o syslog_name=postfix/submission
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
-o smtpd_reject_unlisted_recipient=no
-o smtpd_client_restrictions=permit_sasl_authenticated,reject
-o smtpd_relay_restrictions=permit_sasl_authenticated,reject
```

```
# Choose one: enable submission for loopback clients only, or for a
#127.0.0.1:submission inet n - y - - smtpd
#submission inet n - - y - - smtpd
-o syslog_name=postfix/submission
-o smtpd_tls_security_level=encrypt
-o smtpd_sasl_auth_enable=yes
# -o smtpd_tls_auth_only=yes
# -o local_header_rewrite_clients=static:all
-o smtpd_reject_unlisted_recipient=no
# Instead of specifying complex smtpd_<xxx>_restrictions here,
# specify "smtpd_<xxx>_restrictions=$mua_<xxx>_restrictions"
# here, and specify mua_<xxx>_restrictions in main.cf (where
# "xxx" is "client", "helo", "sender", "relay", or "recipient")
-o smtpd_client_restrictions=
# -o smtpd_helo_restrictions=
# -o smtpd_sender_restrictions=
-o smtpd_relay_restrictions=
# -o smtpd_recipient_restrictions=permit_sasl_authenticated,reject
# -o milter_macro_daemon_name=ORIGINATING
# Choose one: enable submissions for loopback clients only, or for a
```

Figure 7.

Task 2.5: Create Virtual Mailbox Map

- Created the virtual mailbox mapping file:

```
sudo nano /etc/postfix/vmailbox
```

- Added user mappings:

```
user1@example.com example.com/user1/
user2@example.com example.com/user2/
```

- Generated the hash database:

```
sudo postmap /etc/postfix/vmailbox
```

Task 2.6: Create Mail Directory and User

```
sudo groupadd -g 5000 vmail
sudo useradd -g vmail -u 5000 vmail -d /var/mail/vhosts -m
sudo mkdir -p /var/mail/vhosts/example.com
sudo chown -R vmail:vmail /var/mail/vhosts
```


Task 2.7: Restart Postfix

```
sudo systemctl restart postfix
sudo systemctl enable postfix
sudo systemctl status postfix
```

- Deliverable 2.1: Provide the output of `postconf -n` command.
- Deliverable 2.2: Take a screenshot showing Postfix **service status**.

```
root@mail:~# systemctl status postfix
• postfix.service - Postfix Mail Transport Agent
  Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
  Active: active (exited) since Tue 2025-12-09 19:29:00 UTC; 11s ago
  Docs: man:postfix(1)
  Main PID: 2029 (code=exited, status=0/SUCCESS)
  CPU: 4ms

Dec 09 19:29:00 mail.example.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Dec 09 19:29:00 mail.example.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
root@mail:~# postconf -n
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases
append_dot_mydomain = no
biff = no
compatibility_level = 3.6
home_mailbox = Maildir/
inet_interfaces = all
inet_protocols = ipv4
mailbox_size_limit = 0
mydestination = $myhostname, localhost.localdomain, localhost, $mydomain
mydomain = example.com
myhostname = mail.example.com
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
myorigin = $mydomain
readme_directory = no
recipient_delimiter = +
relayhost =
smtp_tls_CApath = /etc/ssl/certs
smtp_tls_security_level = may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache
smtpd_banner = $myhostname ESMTP $mail_name (Ubuntu)
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_type = dovecot
smtpd_tls_cert_file = /etc/ssl/certs/ssl-cert-snakeoil.pem
smtpd_tls_key_file = /etc/ssl/private/ssl-cert-snakeoil.key
smtpd_tls_security_level = may
virtual_gid_maps = static:5000
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_domains = example.com
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
postconf: warning: /etc/postfix/main.cf: unused parameter: smtpd_sasl_auth_clients=yes
```

Figure 8.

Part 3: Dovecot IMAP/POP3 Server Setup

Task 3.1: Install Dovecot

```
sudo apt install -y dovecot-core dovecot-imapd dovecot-pop3d
```

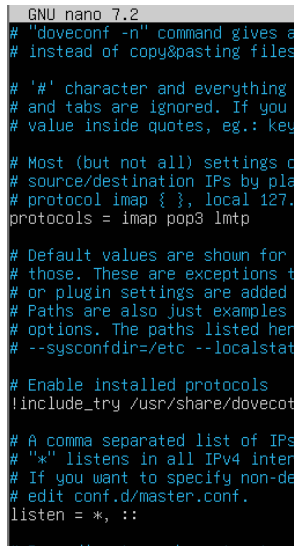
Task 3.2: Configure Dovecot Main Settings

- Edited `/etc/dovecot/dovecot.conf` :

```
sudo nano /etc/dovecot/dovecot.conf
```

- Ensured the following line is uncommented:

```
protocols = imap pop3 lmtp  
listen = *, ::
```



```
GNU nano 7.2  
# "doveconf -n" command gives a  
# instead of copy&pasting files  
  
# '#' character and everything  
# and tabs are ignored. If you  
# value inside quotes, eg.: key  
  
# Most (but not all) settings d  
# source/destination IPs by pla  
# protocol imap { }, local 127.  
protocols = imap pop3 lmtp  
  
# Default values are shown for  
# those. These are exceptions t  
# or plugin settings are added  
# Paths are also just examples  
# options. The paths listed her  
# --sysconfdir=/etc --localstat  
  
# Enable installed protocols  
!include_try /usr/share/dovecot  
  
# A comma separated list of IPs  
# "*" listens in all IPv4 inter  
# If you want to specify non-de  
# edit conf.d/master.conf.  
listen = *, ::
```

Figure 9.

Task 3.3: Configure Mail Location

- Edit `/etc/dovecot/conf.d/10-mail.conf`:

```
sudo nano /etc/dovecot/conf.d/10-mail.conf
```

- Configured mail location:

```
mail_location = maildir:/var/mail/vhosts/%d/%n
mail_privileged_group = mail

namespace inbox {
  inbox = yes
}
```

Command to beatifully display content of the file `grep -v "^#" your_file | grep -v "^$" | less`

- removes the lines starting with "#" and also removes the empty lines, than send the result to `less` for a better display.

```

mail_location = maildir:/var/mail/vhosts/%d/%n
namespace inbox {
    # Namespace type: private, shared or public
    #type = private
    # Hierarchy separator to use. You should use the same separator for all
    # namespaces or some clients get confused. '/' is usually a good one.
    # The default however depends on the underlying mail storage format.
    #separator =
    # Prefix required to access this namespace. This needs to be different for
    # all namespaces. For example "Public/".
    #prefix =
    # Physical location of the mailbox. This is in same format as
    # mail_location, which is also the default for it.
    #location =
    # There can be only one INBOX, and this setting defines which namespace
    # has it.
    inbox = yes
    # If namespace is hidden, it's not advertised to clients via NAMESPACE
    # extension. You'll most likely also want to set list=no. This is mostly
    # useful when converting from another server with different namespaces which
    # you want to deprecate but still keep working. For example you can create
    # hidden namespaces with prefixes "~/mail/", "~%u/mail/" and "mail/".
    #hidden = no
    # Show the mailboxes under this namespace with LIST command. This makes the
    # namespace visible for clients that don't support NAMESPACE extension.
    # "children" value lists child mailboxes, but hides the namespace prefix.
    #list = yes
    # Namespace handles its own subscriptions. If set to "no", the parent
    # namespace handles them (empty prefix should always have this as "yes")
    #subscriptions = yes
    # See 15-mailboxes.conf for definitions of special mailboxes.
}
#type = shared
#separator = /
# Mailboxes are visible under "shared/user@domain/"
# %n, %d and %u are expanded to the destination user.
#prefix = shared/%u/
# Mail location for other users' mailboxes. Note that %variables and ~/
# expands to the logged in user's data. %n, %d, %u and %h expand to the
# destination user's data.
#location = maildir:%h/Maildir:INDEX=~/Maildir/shared/%u
# Use the default namespace for saving subscriptions.
#subscriptions = no
# List the shared/ namespace only if there are visible shared mailboxes.
#list = children
mail_privileged_group = mail
protocol !indexer-worker {
    # If folder vsize calculation requires opening more than this many mails from
    # disk (i.e. mail sizes aren't in cache already), return failure and finish

```

Figure 10.

Task 3.4: Configure Authentication

- Edited `/etc/dovecot/conf.d/10-auth.conf` :

```
sudo nano /etc/dovecot/conf.d/10-auth.conf
```

- Modified the following:

```

disable_plaintext_auth = yes
auth_mechanisms = plain login

```

- Commented out the system auth include and uncommented:

```
!include auth-passwdfile.conf.ext
```

```
#!/include auth-deny.conf.ext
#!/include auth-master.conf.ext

#!/include auth-system.conf.ext
#!/include auth-sql.conf.ext
#!/include auth-ldap.conf.ext
!include auth-passwdfile.conf.ext
#!/include auth-checkpassword.conf.ext
#!/include auth-static.conf.ext
```

Figure 11.

Task 3.5: Configure User Database

- Edited `/etc/dovecot/conf.d/auth-passwdfile.conf.ext` :

```
sudo nano /etc/dovecot/conf.d/auth-passwdfile.conf.ext
```

- Configured:

```
passdb {
  driver = passwd-file
  args = scheme=PLAIN username_format=%u /etc/dovecot/users
}

userdb {
  driver = static
  args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n
}
```

- Created the users file:

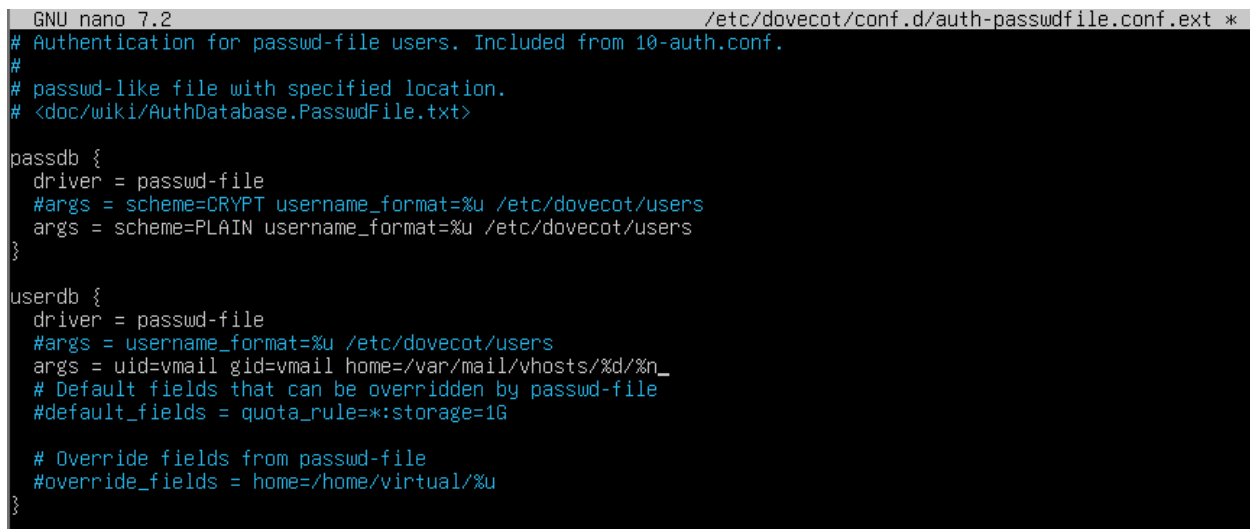
```
sudo nano /etc/dovecot/users
```

- Added users (format: `username:password`):

```
user1@example.com:{PLAIN}password123
user2@example.com:{PLAIN}password456
```

- Set permissions:

```
sudo chmod 640 /etc/dovecot/users
sudo chown root:dovecot /etc/dovecot/users
```



```
GNU nano 7.2 /etc/dovecot/conf.d/auth-passwdfile.conf.ext *
# Authentication for passwd-file users. Included from 10-auth.conf.
#
# passwd-like file with specified location.
# <doc/wiki/AuthDatabase.PasswdFile.txt>
passdb {
  driver = passwd-file
  #args = scheme=CRYPT username_format=%u /etc/dovecot/users
  args = scheme=PLAIN username_format=%u /etc/dovecot/users
}
userdb {
  driver = passwd-file
  #args = username_format=%u /etc/dovecot/users
  args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n_
  # Default fields that can be overridden by passwd-file
  #default_fields = quota_rule=*:storage=1G

  # Override fields from passwd-file
  #override_fields = home=/home/virtual/%u
}
```

Figure 12.

Task 3.6: Configure Dovecot SASL for Postfix

- Edited `/etc/dovecot/conf.d/10-master.conf`:

```
sudo nano /etc/dovecot/conf.d/10-master.conf
```

- Find the service auth section and add:

```
service auth {
  unix_listener /var/spool/postfix/private/auth {
    mode = 0660
```

```
user = postfix
group = postfix
}
}
```

```
service auth {
# auth_socket_path points to this userdb socket by
# used by dovecot-lda, doveadm, possibly imap process
# full permissions to this socket are able to get and
# get the results of everyone's userdb lookups.
#
# The default 0666 mode allows anyone to connect to
# userdb lookups will succeed only if the userdb realm
# matches the caller process's UID. Also if caller's
# socket's uid or gid the lookup succeeds. Anything else
#
# To give the caller full permissions to lookup all
# something else than 0666 and Dovecot lets the kernel
# permissions (e.g. 0777 allows everyone full permissions)
unix_listener auth-userdb {
#mode = 0666
#user =
#group =
}

# Postfix smtp-auth
unix_listener /var/spool/postfix/private/auth {
mode = 0660
user = postfix
group = postfix_
}

# Auth process is run as this user.
#user = $default_internal_user
}
```

Figure 13.

Task 3.7: Restart Dovecot

```
sudo systemctl restart dovecot
sudo systemctl enable dovecot
sudo systemctl status dovecot
```

```

root@mail:~# doveconf -n
# 2.3.21 (47349e2482): /etc/dovecot/dovecot.conf
# Pigeonhole version 0.5.21 (f6cd4b8e)
# OS: Linux 6.8.0-88-generic x86_64 Ubuntu 24.04.3 LTS ext4
# Hostname: mail.example.com
auth_mechanisms = plain login
mail_location = maildir:/var/mail/vhosts/%d/%n
mail_privileged_group = mail
namespace inbox {
  inbox = yes
  location =
  mailbox Drafts {
    special_use = \Drafts
  }
  mailbox Junk {
    special_use = \Junk
  }
  mailbox Sent {
    special_use = \Sent
  }
  mailbox "Sent Messages" {
    special_use = \Sent
  }
  mailbox Trash {
    special_use = \Trash
  }
  prefix =
}
passdb {
  args = scheme=PLAIN username_format=%u /etc/dovecot/users
  driver = passwd-file
}
protocols = imap pop3 lmtp imap lmtp pop3
service auth {
  unix_listener /var/spool/postfix/private/auth {
    group = postfix
    mode = 0660
    user = postfix
  }
}
ssl_cert = </etc/dovecot/private/dovecot.pem
ssl_client_ca_dir = /etc/ssl/certs
ssl_dh = # hidden, use -P to show it
ssl_key = # hidden, use -P to show it
userdb {
  args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n
  driver = passwd-file
}

```

- Deliverable, figure 14: the output of `doveconf -n` command.

```

root@mail:~# ss -tlnp | grep dovecot
LISTEN 0      100        0.0.0.0:995      0.0.0.0:*      users:(("dovecot",pid=2801,fd=24))
LISTEN 0      100        0.0.0.0:993      0.0.0.0:*      users:(("dovecot",pid=2801,fd=42))
LISTEN 0      100        0.0.0.0:110      0.0.0.0:*      users:(("dovecot",pid=2801,fd=22))
LISTEN 0      100        0.0.0.0:143      0.0.0.0:*      users:(("dovecot",pid=2801,fd=40))
LISTEN 0      100        [::]:995        [::]:*         users:(("dovecot",pid=2801,fd=25))
LISTEN 0      100        [::]:993        [::]:*         users:(("dovecot",pid=2801,fd=43))
LISTEN 0      100        [::]:110        [::]:*         users:(("dovecot",pid=2801,fd=23))
LISTEN 0      100        [::]:143        [::]:*         users:(("dovecot",pid=2801,fd=41))

```

- Deliverable, figure 15: Verification that Dovecot is listening on ports 143 and 110 using `ss -tlnp | grep dovecot`.


```

Dec 09 19:45:16 mail.example.com dovecot[2391]: Fatal: service(lmtp) access(/usr/lib/dovecot/lmtp) failed: No such file
Dec 09 19:45:16 mail.example.com dovecot[2391]: master: Fatal: service(lmtp) access(/usr/lib/dovecot/lmtp) failed: No su
Dec 09 19:45:16 mail.example.com systemd[1]: dovecot.service: Main process exited, code=exited, status=89/n/a
Dec 09 19:45:16 mail.example.com systemd[1]: dovecot.service: Failed with result 'exit-code'.
Dec 09 19:45:16 mail.example.com systemd[1]: Failed to start dovecot.service - Dovecot IMAP/POP3 email server.
root@mail:~# systemctl enable dovecot
Synchronizing state of dovecot.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable dovecot
root@mail:~# systemctl status dovecot
* dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
   Active: failed (Result: exit-code) since Tue 2025-12-09 19:45:16 UTC; 28s ago
 Duration: 43min 27.888s
    Docs: man:dovecot(1)
          https://doc.dovecot.org/
 Main PID: 2391 (code=exited, status=89)
    CPU: 18ms

Dec 09 19:45:16 mail.example.com systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
Dec 09 19:45:16 mail.example.com dovecot[2391]: Fatal: service(lmtp) access(/usr/lib/dovecot/lmtp) failed: No such file
Dec 09 19:45:16 mail.example.com dovecot[2391]: master: Fatal: service(lmtp) access(/usr/lib/dovecot/lmtp) failed: No su
Dec 09 19:45:16 mail.example.com systemd[1]: dovecot.service: Main process exited, code=exited, status=89/n/a
Dec 09 19:45:16 mail.example.com systemd[1]: dovecot.service: Failed with result 'exit-code'.
Dec 09 19:45:16 mail.example.com systemd[1]: Failed to start dovecot.service - Dovecot IMAP/POP3 email server.
root@mail:~# systemctl status dovecot^C
root@mail:~# ^C
root@mail:~# sudo apt install dovecot-lmtpd

```

Figure 16. Status of services (error is present)

Got error that dovecot missed `lmtp` that we configured inside `/etc/dovecot/dovecot.conf`, so I installed this package for dovecot and restarted. Error disappeared.

```

root@mail:~# systemctl status dovecot
• dovecot.service - Dovecot IMAP/POP3 email server
   Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
   Active: active (running) since Tue 2025-12-09 19:46:59 UTC; 4s ago
     Docs: man:dovecot(1)
           https://doc.dovecot.org/
 Main PID: 2801 (dovecot)
   Status: "v2.3.21 (47349e2482) running"
    Tasks: 4 (limit: 7023)
  Memory: 3.5M (peak: 3.8M)
     CPU: 62ms
   CGroup: /system.slice/dovecot.service
           └─2801 /usr/sbin/dovecot -F
             └─2803 dovecot/anvil
               └─2804 dovecot/log
                 └─2805 dovecot/config

Dec 09 19:46:59 mail.example.com systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
Dec 09 19:46:59 mail.example.com dovecot[2801]: master: Dovecot v2.3.21 (47349e2482) starting up for imap, pop3, lmtp, .
Dec 09 19:46:59 mail.example.com systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email server.

```

Figure 17. Updated status of dovecot

Part 4: Security Implementation - TLS/SSL Encryption

Task 4.1: Generate Self-Signed SSL Certificate (for testing)

```
sudo mkdir -p /etc/ssl/mail
cd /etc/ssl/mail
sudo openssl req -new -x509 -days 365 -nodes \
-out mail.example.com.crt \
-keyout mail.example.com.key
```

- Fill in the certificate details when prompted.



BLA-BLA certificate details may cause errors in identifying the certificate.

- Set proper permissions:

```
sudo chmod 600 /etc/ssl/mail/mail.example.com.key
sudo chmod 644 /etc/ssl/mail/mail.example.com.crt
```

```
root@mail:~# sudo mkdir -p /etc/ssl/mail  
root@mail:~# cd /etc/ssl/mail/  
root@mail:/etc/ssl/mail# sudo openssl req -new -x509 -days 365 -nodes -out mail.example.com.crt -keyout mail.example.com.key  
.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+.+  
-----  
You are about to be asked to enter information that will be incorporated  
into your certificate request.  
  
What you are about to enter is what is called a Distinguished Name or a DN.  
There are quite a few fields but you can leave some blank  
For some fields there will be a default value,  
If you enter '.', the field will be left blank.  
-----  
Country Name (2 letter code) [AU]:RU  
State or Province Name (full name) [Some-State]:  
Locality Name (eg, city) []:  
Organization Name (eg, company) [Internet Widgits Pty Ltd]:Quantet  
Organizational Unit Name (eg, section) []:  
Common Name (e.g., server FQDN or YOUR name) []:Email server  
Email Address []:user@example.com  
root@mail:/etc/ssl/mail# sudo chmod 600 /etc/ssl/mail/mail.example.com.key  
root@mail:/etc/ssl/mail# sudo chmod 644 /etc/ssl/mail/mail.example.com.crt
```

Figure 18.

```

Requesting a certificate for mail.example.com
An unexpected error occurred:
Error creating new order :: Cannot issue for "mail.example.com": The ACME server refuses to issue a certificate for this domain name
policy
Ask for help or search for solutions at https://community.letsencrypt.org. See the logfile /var/log/letsencrypt/letsencrypt.log or
re details.
root@mail:/etc/ssl/mail# certbot certonly --standalone -d mail.example.com

```

Figure 19.

For a local domain, it is not possible to create a certificate with Let's Encrypt, because this type of domain is in the block list.

Task 4.2: Configure Postfix TLS

- Edited `/etc/postfix/main.cf` and add:

```
# TLS Settings
smtpd_tls_cert_file=/etc/ssl/mail/mail.example.com.crt
smtpd_tls_key_file=/etc/ssl/mail/mail.example.com.key
smtpd_use_tls=yes
smtpd_tls_security_level=may
smtpd_tls_auth_only = yes
smtpd_tls_loglevel = 1
smtpd_tls_received_header = yes
smtpd_tls_session_cache_timeout = 3600s

# Outgoing TLS
smtp_tls_security_level = may
smtp_tls_loglevel = 1
```

Task 4.3: Configure Dovecot SSL

- Edited `/etc/dovecot/conf.d/10-ssl.conf` :

```
sudo nano /etc/dovecot/conf.d/10-ssl.conf
```

- Configured SSL settings:

```
ssl = required
ssl_cert = </etc/ssl/mail/mail.example.com.crt
ssl_key = </etc/ssl/mail/mail.example.com.key
ssl_min_protocol = TLSv1.2
ssl_prefer_server_ciphers = yes
```

Task 4.4: Restart Services

```
sudo systemctl restart postfix
sudo systemctl restart dovecot
```

```
root@mail:/etc/ssl/mail# systemctl restart postfix
root@mail:/etc/ssl/mail# systemctl restart dovecot
root@mail:/etc/ssl/mail# systemctl status dovecot
• dovecot.service - Dovecot IMAP/POP3 email server
  Loaded: loaded (/usr/lib/systemd/system/dovecot.service; enabled; preset: enabled)
  Active: active (running) since Tue 2025-12-09 22:24:37 UTC; 6s ago
    Docs: man:dovecot(1)
          https://doc.dovecot.org/
  Main PID: 3553 (dovecot)
  Status: "v2.3.21 (47349e2482) running"
    Tasks: 4 (limit: 7023)
  Memory: 3.5M (peak: 3.9M)
    CPU: 46ms
  CGroup: /system.slice/dovecot.service
          └─3553 /usr/sbin/dovecot -F
             3554 dovecot/anvil
             3555 dovecot/log
             3556 dovecot/config

Dec 09 22:24:37 mail.example.com systemd[1]: Starting dovecot.service - Dovecot IMAP/POP3 email server...
Dec 09 22:24:37 mail.example.com dovecot[3553]: master: Dovecot v2.3.21 (47349e2482) starting up for imap
Dec 09 22:24:37 mail.example.com systemd[1]: Started dovecot.service - Dovecot IMAP/POP3 email server.
root@mail:/etc/ssl/mail# systemctl status postfix
• postfix.service - Postfix Mail Transport Agent
  Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
  Active: active (exited) since Tue 2025-12-09 22:24:30 UTC; 19s ago
    Docs: man:postfix(1)
  Process: 3542 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
  Main PID: 3542 (code=exited, status=0/SUCCESS)
    CPU: 5ms

Dec 09 22:24:30 mail.example.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent...
Dec 09 22:24:30 mail.example.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent.
```

Figure 20.

Task 4.5: Configure Firewall

- Allow necessary mail server ports:

```
sudo ufw allow 25/tcp
sudo ufw allow 587/tcp
sudo ufw allow 465/tcp
sudo ufw allow 143/tcp
sudo ufw allow 993/tcp
sudo ufw allow 110/tcp
sudo ufw allow 995/tcp
sudo ufw allow 80/tcp
sudo ufw allow 443/tcp
sudo ufw enable
```

OR BETTER OPTION

for p in 25 587 465 143 993 110 995 80 443; do sudo ufw allow \${p}/tcp; done && sudo ufw enable

```
root@mail:/etc/ssl/mail# for p in 25 587 465 143 993 110 995 80 443; do sudo ufw allow ${p}/tcp; done && sudo ufw enable
Skipping adding existing rule
Skipping adding existing rule (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Rules updated
Rules updated (v6)
Skipping adding existing rule
Skipping adding existing rule (v6)
Rules updated
Rules updated (v6)
Firewall is active and enabled on system startup
root@mail:/etc/ssl/mail#
```

- Deliverable, figure 21: firewall rules updated

- Test of TLS connection using: `openssl s_client -connect localhost:587 -starttls smtp`

Here, I ran into the issue - can't connect to the email. So I executed `ss -tlnp | grep 587` and saw nothing! That means I didn't turn on something. I checked master.cf and observed that I had not commented out one specific line for submission.

Was:

```
#submission inet n      -       -       -       smtpd
```

Become:

```
submission inet n      -       -       -       smtpd
```

```

GNU nano 7.2 /etc/postfix/master.cf *
#
# Postfix master process configuration file. For details on the format
# of the file, see the master(5) manual page (command: "man 5 master" or
# on-line: http://www.postfix.org/master.5.html).
#
# Do not forget to execute "postfix reload" after editing this file.
#
# =====
# service type private unpriv chroot wakeup maxproc command + args
#          (yes)   (yes)   (no)   (never) (100)
# =====
smtp      inet  n       -       y       -       -       smtpd
#smtp     inet  n       -       y       -       1       postscreen
#smtpd    pass  -       -       y       -       -       smtpd
#dnsblog  unix  -       -       y       -       0       dnsblog
#tlsproxy unix  -       -       y       -       0       tlsproxy
# Choose one: enable submission for loopback clients only, or for any client.
#127.0.0.1:submission inet n - y - - smtpd
submission inet n       -       y       -       -       smtpd
  -o syslog_name=postfix/submission
  -o smtpd_tls_security_level=encrypt
  -o smtpd_sasl_auth_enable=yes
#  -o smtpd_tls_auth_only=yes
#  -o local_header_rewrite_clients=static:all
#  -o smtpd_reject_unlisted_recipient=no
#    Instead of specifying complex smtpd_<xxx>_restrictions here,
#    specify "smtpd_<xxx>_restrictions=$mua_<xxx>_restrictions"
#    here, and specify mua_<xxx>_restrictions in main.cf (where
#    "<xxx>" is "client", "helo", "sender", "relay", or "recipient").
#  -o smtpd_client_restrictions=
#  -o smtpd_helo_restrictions=
#  -o smtpd_sender_restrictions=
#  -o smtpd_relay_restrictions=
#  -o smtpd_recipient_restrictions=permit_sasl_authenticated,reject

```

Figure 22. Misconfiguration fixed

```

Email Server [Running] - Oracle VirtualBox
File Machine View Input Devices Help

Server Temp Key: X25519, 253 bits
---
SSL handshake has read 1757 bytes and written 406 bytes
Verification error: self-signed certificate
---
New, TLSv1.3, Cipher is TLS_AES_256_GCM_SHA384
Server public key is 2048 bit
Secure Renegotiation IS NOT supported
Compression: NONE
Expansion: NONE
No ALPN negotiated
Early data was not sent
Verify return code: 18 (self-signed certificate)
---
250 CHUNKING
---
Post-Handshake New Session Ticket arrived:
SSL-Session:
    Protocol : TLSv1.3
    Cipher : TLS_AES_256_GCM_SHA384
    Session-ID: AC070F01E703516A9DE4291EBC20426A279811E62A6CC2B7F812911EA3EE0652
    Session-ID-ctx:
    Resumption PSK: 6CEC0240CBB9319A317E10A3B9CC2B3830C27240B76537BE0C1702F742118AFC133215325DC787805A6C0F
    PSK identity: None
    PSK identity hint: None
    SRP username: None
    TLS session ticket lifetime hint: 7200 (seconds)
    TLS session ticket:
    0000 - f1 5f 0b b2 ba e1 c4 27-ba 14 98 f5 f9 a5 f9 71 .....0
    0010 - de 94 46 9f de ae 30 10-41 36 1f f4 18 9b 6a 55 ..F...0.A6....JU
    0020 - 9f eb 80 5a 6c 4c 6b 6a-b9 1f 9a 10 6d 1c 01 e7 ...ZlUkJ....im..
    0030 - dd 9b b0 a6 23 97 43 c3-64 29 47 6b 8a 19 23 ba ...#.C.d)Gk..#.
    0040 - ea 88 87 b2 f1 29 e5 ff-df 44 6e 8b fa 56 a6 59 .....Dn..V.Y
    0050 - 7f 0f a1 35 94 41 76 40-87 e5 e5 98 91 b3 b5 dc ...5.Av0.....
    0060 - 98 f1 0a aa a5 2a 42 6f-b1 73 64 5e 09 22 a7 59 .....*8o,sd...Y
    0070 - b8 93 68 58 2d d5 86 62-53 6f 3f b0 b1 4c 5b 1c ..hX-..bSo?...L[
    0080 - 30 73 1e 29 2e f4 b3 96-e4 84 f4 cd 63 ba 4c 44 0s.).....c.LD
    0090 - 12 3a ad a0 56 c9 b2 b5-42 f0 16 ec c6 ce 14 cc ...V...B.....
    00a0 - 7d ff 6b 1c 62 67 37 c6-83 90 b8 b1 5a 6d 13 7a }.k.bg7.....Zm.z
    00b0 - 08 c4 71 c2 b4 da d0 e2-96 30 10 62 bc df 87 5e ..q.....0.b...
    00c0 - e3 26 59 55 bb dc 15 db-0c a5 1f ab 92 0e 82 c3 ..&YU.....

    Start Time: 1765319776
    Timeout : 7200 (sec)
    Verify return code: 18 (self-signed certificate)
    Extended master secret: no
    Max Early Data: 0
---
read R BLOCK

```

Figure 23. Successfully connected using TLS 1.3

```

root@mail:/share# openssl s_client -connect localhost:25 -starttls smtp > tls_test_output.txt
Can't use SSL_get_servername
depth=0 C = RU, ST = Some-State, O = Quartet, CN = Email server, emailAddress = user1@example.com
verify error:num=18:self-signed certificate
verify return:1
depth=0 C = RU, ST = Some-State, O = Quartet, CN = Email server, emailAddress = user1@example.com
verify return:1
250 CHUNKING

```

Figure 24. Successfully copied connection log to shared folder for the report.

Part 5: Email Authentication Configuration

Task 5.2: Install OpenDKIM

```
sudo apt install -y opendkim opendkim-tools
```

Task 5.3: Configure OpenDKIM

- Edited `/etc/opendkim.conf` :

```
sudo nano /etc/opendkim.conf
```

- Configured:

Domain	example.com
KeyFile	/etc/opendkim/keys/example.com/default.private
Selector	default
Socket	inet:8891@localhost
Mode	sv
Syslog	yes
LogWhy	yes
Canonicalization	relaxed/simple


```

GNU nano 7.2 /etc/openssl.conf
# This is a basic configuration for signing and verifying. It can easily be
# adapted to suit a basic installation. See openssl.conf(5) and
# /usr/share/doc/openssl/examples/openssl.conf.sample for complete
# documentation of available configuration parameters.

Syslog                yes
SyslogSuccess          yes
LogWhy                 yes

# Common signing and verification parameters. In Debian, the "From" header is
# oversigned, because it is often the identity key used by reputation systems
# and thus somewhat security sensitive.
Canonicalization      relaxed/simple
Mode                  sv
#SubDomains            no
OversignHeaders        From

# Signing domain, selector, and key (required). For example, perform signing
# for domain "example.com" with selector "2020" (2020._domainkey.example.com),
# using the private key stored in /etc/dkimkeys/example.private. More granular
# setup options can be found in /usr/share/doc/openssl/README.openssl.
Domain                example.com
Selector              default
KeyFile               /etc/openssl/keys/example.com/default.private

# In Debian, openssl runs as user "openssl". A umask of 007 is required when
# using a local socket with MTAs that access the socket as a non-privileged
# user (for example, Postfix). You may need to add user "postfix" to group
# "openssl" in that case.
UserID                openssl
UMask                 007

# Socket for the MTA connection (required). If the MTA is inside a chroot jail,
# it must be ensured that the socket is accessible. In Debian, Postfix runs in
# a chroot in /var/spool/postfix, therefore a Unix socket would have to be
# configured as shown on the last line below.
#Socket               local:/run/openssl/openssl.sock
Socket                inet:8891@localhost
#Socket               inet:8891
#Socket               local:/var/spool/postfix/openssl/openssl.sock

PidFile               /run/openssl/openssl.pid

# Hosts for which to sign rather than verify, default is 127.0.0.1. See the
# OPERATION section of openssl(8) for more information.
#InternalHosts        192.168.0.0/16, 10.0.0.0/8, 172.16.0.0/12

```

Figure 25.

Task 5.4: Generate DKIM Keys

```

sudo mkdir -p /etc/openssl/keys/example.com
sudo openssl-genkey -D /etc/openssl/keys/example.com -d example.com
-s default
sudo chown -R openssl:openssl /etc/openssl/keys/

```

- View the public key:

```
sudo cat /etc/opendkim/keys/example.com/default.txt
```

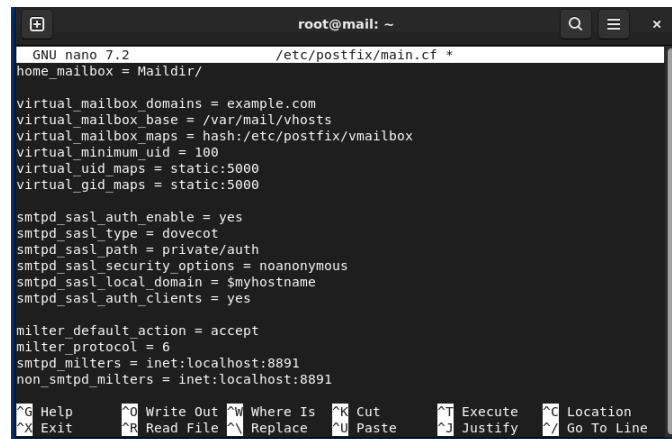
```
root@mail:~/share# mkdir -p /etc/opendkim/keys/example.com
root@mail:~/share# opendkim-genkey -D /etc/opendkim/keys/example.com -d example.com -s def[ 6519.364122] U
/linux-6.8.0/drivers/virt/vboxguest/vboxguest_utils.c:367:20
[ 6519.364629] index 1 is out of range for type '___u64 [1]'
^C
root@mail:~/share# opendkim-genkey -D /etc/opendkim/keys/example.com -d example.com -s default
root@mail:~/share# chown -R opendkim:opendkim /etc/opendkim/keys/
root@mail:~/share# cat /etc/opendkim/keys/example.com/default.txt
default._domainkey IN TXT ("v=DKIM1; h=sha256; k=rsa; "
"p=MIIBIjANBgkqhkiG9w0BAQEFAAOCAQ8AMIIBCgKCAQEAyBPPFJJZQZw/D3u0s+0/wb5ZcnLunKY3bsK9G0Qa8G1d5oyf
c0if1D0yAHYHYDFFzL7MU1efQVLPOGdKPMJK1uuMgaP8iug83cs1U/b2ANILKxQuWTVoMEMkbhg48h26kLJQFMDNUSP1UCMeEah13o"
"A9FQRecw/pMuspad9qa9/v9KxNoNKJ4qdXr3/jn9jb64LMf9tiHp/kZuJ3HdMISVSSc0fSm0uHKCdRRsYu0q4J6ck6IuVyu
----- DKIM key default for example.com
```

Figure 26.

Task 5.6: Connect OpenDKIM to Postfix

- Edited `/etc/postfix/main.cf` and add:

```
# DKIM
milter_default_action = accept
milter_protocol = 6
smtpd_milters = inet:localhost:8891
non_smtpd_milters = inet:localhost:8891
```



```
root@mail: ~
GNU nano 7.2 /etc/postfix/main.cf *
home_mailbox = Maildir/

virtual_mailbox_domains = example.com
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
virtual_gid_maps = static:5000

smtpd_sasl_auth_enable = yes
smtpd_sasl_type = dovecot
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_auth_clients = yes

milter_default_action = accept
milter_protocol = 6
smtpd_milters = inet:localhost:8891
non_smtpd_milters = inet:localhost:8891

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^M Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

Figure 27.

Task 5.8: Restart Services

```
sudo systemctl restart opendkim
sudo systemctl restart postfix
```

- Deliverable 5.1: Provide the DKIM public key from default.txt.
- Deliverable 5.2: Show OpenDKIM service status.

```
root@mail:~# systemctl status postfix
● postfix.service - Postfix Mail Transport Agent
   Loaded: loaded (/usr/lib/systemd/system/postfix.service; enabled; preset: enabled)
   Active: active (exited) since Wed 2025-12-10 15:45:47 UTC; 1min 47s ago
     Docs: man:postfix(1)
   Process: 4378 ExecStart=/bin/true (code=exited, status=0/SUCCESS)
    Main PID: 4378 (code=exited, status=0/SUCCESS)
      CPU: 2ms

Dec 10 15:45:47 mail.example.com systemd[1]: Starting postfix.service - Postfix Mail Transport Agent:
Dec 10 15:45:47 mail.example.com systemd[1]: Finished postfix.service - Postfix Mail Transport Agent:
lines 1-10/10 (END)

systemctl status opendkim
● opendkim.service - OpenDKIM Milter
   Loaded: loaded (/usr/lib/systemd/system/opendkim.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-12-10 15:45:42 UTC; 2min 7s ago
     Docs: man:opendkim(8)
           man:opendkim.conf(5)
           man:opendkim_lua(3)
           man:opendkim_genkey(8)
           man:opendkim_genzone(8)
           man:opendkim_testkey(8)
           http://www.opendkim.org/docs.html
   Process: 4014 ExecStart=/usr/sbin/opendkim (code=exited, status=0/SUCCESS)
    Main PID: 4017 (opendkim)
      Tasks: 6 (limit: 9434)
     Memory: 2.0M (peak: 2.5M)
        CPU: 17ms
    CGroup: /system.slice/opendkim.service
            └─4017 /usr/sbin/opendkim

Dec 10 15:45:42 mail.example.com systemd[1]: Starting opendkim.service - OpenDKIM Milter...
Dec 10 15:45:42 mail.example.com systemd[1]: Started opendkim.service - OpenDKIM Milter.
Dec 10 15:45:42 mail.example.com opendkim[4017]: OpenDKIM Filter v2.11.0 starting
```

Figure 28.

Part 6: Webmail Interface Setup (Roundcube)

Task 6.1: Install Web Server and PHP

```
sudo apt install -y nginx php-fpm php-cli php-imap php-json \\\n    php-mysql php-mbstring php-xml php-zip php-intl \\\n    php-imagick php-curl php-gd
```

Task 6.2: Install MariaDB

```
sudo apt install -y mariadb-server
```

```
sudo mysql_secure_installation
```

Task 6.3: Create Database for Roundcube

```
sudo mysql -u root -p
```

- Run the following SQL commands:

```
CREATE DATABASE roundcube CHARACTER SET utf8mb4 COLLATE utf8mb4_
unicode_ci;
CREATE USER 'roundcube'@'localhost' IDENTIFIED BY 'strong_password';
GRANT ALL PRIVILEGES ON roundcube.* TO 'roundcube'@'localhost';
FLUSH PRIVILEGES;
EXIT;
```

Task 6.4: Download and Install Roundcube

```
cd /tmp
wget https://github.com/roundcube/roundcubemail/releases/download/1.6.5/r
oundcubemail-1.6.5-complete.tar.gz
tar -xzf roundcubemail-1.6.5-complete.tar.gz
sudo mv roundcubemail-1.6.5 /var/www/roundcube
sudo chown -R www-data:www-data /var/www/roundcube
```



```
roundcubemail-1.6.5-complete. 100%[+++++>] 5.75M 1.05MB/s in 4.5s
2025-12-10 16:01:33 (1.05 MB/s) - 'roundcubemail-1.6.5-complete.tar.gz' saved [6028873/6028873]
root@mail:/tmp# tar -xzf roundcubemail-1.6.5-complete.tar.gz
root@mail:/tmp# mv roundcubemail-1.6.5 /var/www/roundcube
root@mail:/tmp# chown -R www-data:www-data /var/www/roundcube/
```

Figure 29.

Here I have error I can't access login page after installation.

Firstly, I found that I did not put right password for roundcube user, then I reinstalled it and put it right. But error is the same. Then I tried accessing mysql

via roundcube user with the password - good.

I opened logs folder inside `var/www/roundcube` and saw errors:

Error: [1146] Table 'roundcube.session' doesn't exist

Now I manually add this schema (This imports the initial tables (including 'session') from Roundcube's SQL file):

```
mysql -u roundcube -pstrong_password roundcube < /var/www/roundcube/SQL/mysql.initial.sql
```

```
root@mail:/var/www/roundcube/logs# mysql -u roundcube -pstrong_password roundcube < /var/www/roundcube/SQL/mysql.initial.sql
root@mail:/var/www/roundcube/logs# systemctl restart nginx mariadb php8.3-fpm
root@mail:/var/www/roundcube/logs#
```

Figure 29. Restart of main services

Error is solved. GPT helped with adding the schema, since the schema wasn't imported.

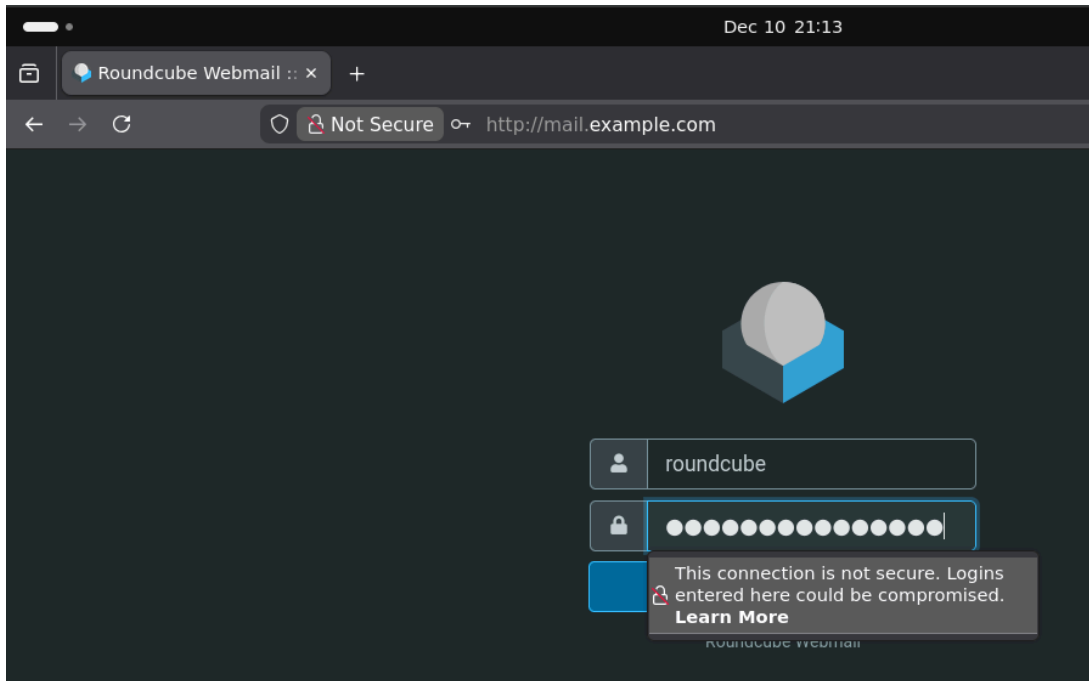


Figure 30. Login page of roundcube

Task 6.5: Configure Nginx for Roundcube

- Create Nginx configuration:

```
sudo nano /etc/nginx/sites-available/roundcube
```

- Add:

```
server {  
    listen 80;  
    server_name mail.example.com;  
    root /var/www/roundcube;  
    index index.php;  
  
    location / {  
        try_files $uri $uri/ /index.php;  
    }  
  
    location ~ \.php$ {
```

```

    include snippets/fastcgi-php.conf;
    fastcgi_pass unix:/var/run/php/php8.1-fpm.sock;
}

location ~ /\.ht {
    deny all;
}
}

```

- Enable the site:

```

sudo ln -s /etc/nginx/sites-available/roundcube /etc/nginx/sites-enabled/
sudo nginx -t
sudo systemctl restart nginx

```

```

root@mail: /tmp
GNU nano 7.2 /etc/nginx/sites-available/roundcube *
server {
    listen 80;
    server_name mail.example.com;
    root /var/www/roundcube;
    index index.php;

    location / {
        try_files $uri $uri/ /index.php;
    }

    location ~ \.php$ {
        include snippets/fastcgi-php.conf;
        fastcgi_pass unix:/var/run/php/php8.1-fpm.sock;
    }

    location ~ /\.ht {
        deny all;
    }
}

```

Figure 31. Nginx configuration (with a syntax error and old socket version)

```

root@mail:/tmp# nginx -t
2025/12/10 16:09:56 [warn] 4584#4584: server name "/var/www/roundcube" has suspicious symbols in /etc/nginx/sites-enabled/roundcube:4
nginx: the configuration file /etc/nginx/nginx.conf syntax is ok
nginx: configuration file /etc/nginx/nginx.conf test is successful
root@mail:/tmp# systemctl restart nginx
root@mail:/tmp# systemctl status nginx
● nginx.service - A high performance web server and a reverse proxy server
   Loaded: loaded (/usr/lib/systemd/system/nginx.service; enabled; preset: enabled)
   Active: active (running) since Wed 2025-12-10 16:10:31 UTC; 3s ago
     Docs: man:nginx(8)
   Process: 4593 ExecStartPre=/usr/sbin/nginx -t -q -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Process: 4597 ExecStart=/usr/sbin/nginx -g daemon on; master_process on; (code=exited, status=0/SUCCESS)
   Main PID: 4598 (nginx)
    Tasks: 4 (limit: 9434)
   Memory: 3.1M (peak: 3.2M)
      CPU: 20ms
   CGroup: /system.slice/nginx.service
           └─4598 "nginx: master process /usr/sbin/nginx -g daemon on; master_process on;"
             └─4599 "nginx: worker process"
               └─4600 "nginx: worker process"
                 └─4601 "nginx: worker process"

Dec 10 16:10:31 mail.example.com systemd[1]: Starting nginx.service - A high performance web server and a reverse proxy
Dec 10 16:10:31 mail.example.com nginx[4593]: 2025/12/10 16:10:31 [warn] 4593#4593: server name "/var/www/roundcube" has suspicious symbols in /etc/nginx/sites-enabled/roundcube:4
Dec 10 16:10:31 mail.example.com nginx[4597]: 2025/12/10 16:10:31 [warn] 4597#4597: server name "/var/www/roundcube" has suspicious symbols in /etc/nginx/sites-enabled/roundcube:4
Dec 10 16:10:31 mail.example.com systemd[1]: Started nginx.service - A high performance web server and a reverse proxy

```

Figure 32. nginx service status

Task 6.6: Configure Roundcube

- Visit <http://mail.example.com/installer> in your browser to complete the installation.
- Database setup:
- Database type: MySQL
- Database name: roundcube
- Database user: roundcube
- Database password: (your password)
- IMAP settings:
- IMAP server: ssl://localhost:993
- SMTP settings:
- SMTP server: tls://localhost:587
- Use authentication: Yes
- After configuration, download the config.inc.php file and place it in `/var/www/roundcube/config/`.
- Remove the installer for security:


```
sudo rm -rf /var/www/roundcube/installer
```

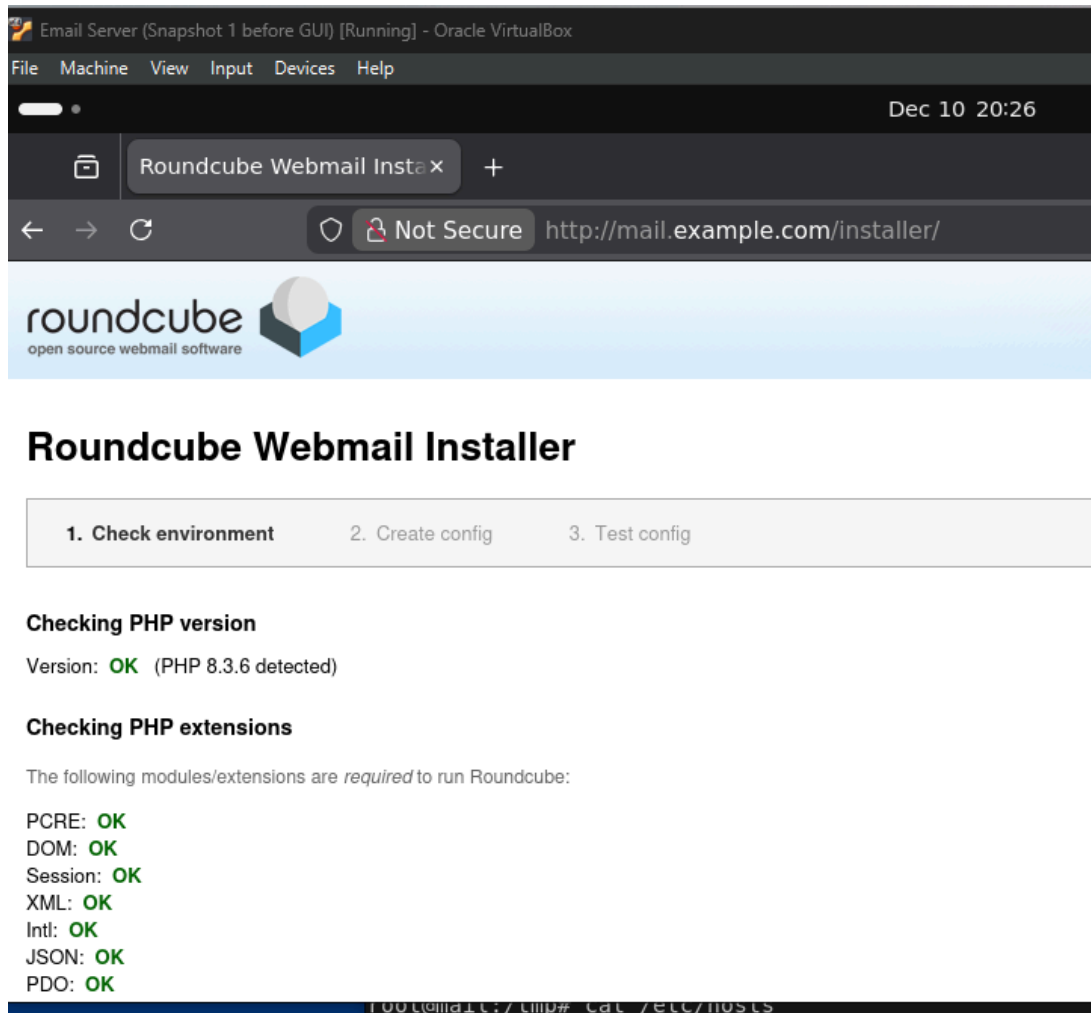


Figure 33. a screenshot of the Roundcube installation page



Figure 34. A screenshot of the Roundcube installation page (2)



I had problems deploying the site: default nginx settings were disrupting me. So I fully reinstalled nginx, deleted all folders and reinstalled roundcube and rewrote nginx conf for the site.

Also I changed fpm php socket to use 8.3 version, because 8.1 is not supported.

Part 7: Testing and Troubleshooting

Task 7.1: Test SMTP with Telnet

- Test basic SMTP connectivity:

```
telnet localhost 25
```

- Commands to execute:

```
EHLO localhost
MAIL FROM: <user1@example.com>
RCPT TO: <user2@example.com>
DATA
Subject: Test Email
This is a test email.
.
QUIT
```

```
RCPT TO: user2@example.com
501 5.5.4 Syntax: RCPT TO:<address>
RCPT TO:user2@example.com
550 5.1.1 <user2@example.com>: Recipient address rejected: User unknown in local recipient table
```

Figure 35. Error about `user2`

The error "550 5.1.1 `user2@example.com`: Recipient address rejected: User unknown in local recipient table" occurs because Postfix is treating "`example.com`" as a local domain instead of a virtual domain. As a result, it's checking for "user2" as a system user (e.g., in `/etc/passwd`), which doesn't exist

In my Postfix configuration (`/etc/postfix/main.cf`) I had:

```
mydestination = $myhostname, localhost.$mydomain, localhost, $mydomain
```

So I changed to:

```
mydestination = $myhostname, localhost.$mydomain, localhost
```

And now IMAP and SMTP tests work correctly.

```
MAIL FROM:user1@example.com
250 2.1.0 Ok
RCPT TO:user2@example.com
250 2.1.5 Ok
DATA
354 End data with <CR><LF>.<CR><LF>
Subject: Orange tree
This is dungeon with apples.
.
250 2.0.0 Ok: queued as 6685C141129
QUIT
221 2.0.0 Bye
Connection closed by foreign host.
root@mail:~# echo "Test" | mail -s "Test subject" user2@example.com
root@mail:~# sudo tail -f /var/log/mail.log
2025-12-10T21:31:33.964124+00:00 mail postfix/qmgr[4913]: 6685C141129: removed
2025-12-10T21:31:35.667872+00:00 mail postfix/smtpd[4921]: disconnect from localhost[127.0.0.1] ehlo=1 mail=1 rcpt=1 data=1 quit=1 commands=5
2025-12-10T21:31:51.217440+00:00 mail postfix/pickup[4912]: 34DE81422FC: uid=0 from=<root@mail.example.com>
2025-12-10T21:31:51.221105+00:00 mail postfix/cleanup[4930]: 34DE81422FC: message-id=<20251210213151.34DE81422FC@mail.example.com>
2025-12-10T21:31:51.221644+00:00 mail opendkim[1229]: 34DE81422FC: no signing domain match for 'mail.example.com'
2025-12-10T21:31:51.221689+00:00 mail opendkim[1229]: 34DE81422FC: no signing subdomain match for 'mail.example.com'
2025-12-10T21:31:51.222309+00:00 mail opendkim[1229]: 34DE81422FC: no signature data
2025-12-10T21:31:51.236755+00:00 mail postfix/qmgr[4913]: 34DE81422FC: from=<root@mail.example.com>, size=375, nrcpt=1 (queue active)
2025-12-10T21:31:51.245253+00:00 mail postfix/virtual[4939]: 34DE81422FC: to=<user2@example.com>, relay=virtual, delay=0.09, delays=0.08/0/0/0.01, dsn=2.0.0, status=sent (delivered to maildir)
2025-12-10T21:31:51.245331+00:00 mail postfix/qmgr[4913]: 34DE81422FC: removed
```

Figure 36. Pipeline of IMAP/SMTP tests and content of the `mail.log`

Task 7.2: Test SMTP with TLS

```
openssl s_client -connect localhost:587 -starttls smtp
```

- After connection, test SMTP commands.

Task 7.3: Test IMAP Connection

```
openssl s_client -connect localhost:993
```

- Commands:

```
a001 LOGIN user1@example.com password123
a002 LIST "" "*"
a003 SELECT INBOX
a004 LOGOUT
```

Here I was not able to normally run commands so -quiet option helped to calm down SSL warnings (to suppress certificate warnings).

```
openssl s_client -connect localhost:993 -quiet
```

Task 7.4: Send Test Email

```
echo "Test email body" | mail -s "Test Subject" user2@example.com
```

Task 7.5: Check Mail Logs

- View Postfix logs:

```
sudo tail -f /var/log/mail.log
```

- Or on some systems:

```
sudo tail -f /var/log/maillog
```

Task 7.6: Verify Email Authentication

- Send an email to an external address (Gmail, etc.) and check headers for:
- SPF: PASS
- DKIM: PASS
- DMARC: PASS

```
2025-12-10T21:43:59.352008+00:00 mail postfix/qmgr[14913]: 51AEA1422FD: from=<root@mail.example.com>, size=382, in-cpt=1 (queue active)
2025-12-10T21:44:20.865072+00:00 mail postfix/smtp[5184]: connect to gmail-smtp-in.l.google.com[142.250.102.26]:25: Connection refused
```

Figure 37. Log of sending email to an external address (Screenshot showing email headers with SPF, DKIM, and DMARC results.)

Because this is local domain and email, we cannot send emails to internet, only inside LAN.

Task 7.7: Test Mail Queue

- Check mail queue:

```
mailq
postqueue -p
```

- Flush queue:

```
postqueue -f
```

```
root@mail:~# mailq
-Queue ID- --Size-- ----Arrival Time---- -Sender/Recipient-----
51AEA1422FD      382 Wed Dec 10 21:43:59  root@mail.example.com
(connect to alt4.gmail-smtp-in.l.google.com[108.177.125.27]:25: Connection refused)
ktofreesapiens@gmail.com

-- 0 Kbytes in 1 Request.
```

Figure 38. Error on sending email to an external address (Screenshot showing email headers with SPF, DKIM, and DMARC results.)

Task 7.8: Common Troubleshooting Commands

- Check Postfix configuration:

```
postconf -n
```



```
root@mail:~# postconf -n
alias_database = hash:/etc/aliases
alias_maps = hash:/etc/aliases
append_dot_mydomain = no
biff = no
compatibility_level = 3.6
home_mailbox = Maildir/
inet_interfaces = all
inet_protocols = ipv4
mailbox_size_limit = 0
milter_default_action = accept
milter_protocol = 6
mydestination = $myhostname, localhost.localdomain, localhost
mydomain = example.com
myhostname = mail.example.com
mynetworks = 127.0.0.0/8 [::ffff:127.0.0.0]/104 [::1]/128
myorigin = $mydomain
non_smtpd_milters = inet:localhost:8891
readme_directory = no
recipient_delimiter = +
relayhost =
smtp_tls_CApath = /etc/ssl/certs
smtp_tls_security_level = may
smtp_tls_session_cache_database = btree:${data_directory}/smtp_scache
smtpd_banner = $myhostname ESMTPEX $mail_name (Ubuntu)
smtpd_milters = inet:localhost:8891
smtpd_relay_restrictions = permit_mynetworks permit_sasl_authenticated defer_unauth_destination
smtpd_sasl_auth_enable = yes
smtpd_sasl_local_domain = $myhostname
smtpd_sasl_path = private/auth
smtpd_sasl_security_options = noanonymous
smtpd_sasl_type = dovecot
smtpd_tls_auth_only = yes
smtpd_tls_cert_file = /etc/ssl/mail/mail.example.com.crt
smtpd_tls_key_file = /etc/ssl/mail/mail.example.com.key
smtpd_tls_loglevel = 1
smtpd_tls_received_header = yes
smtpd_tls_security_level = may
smtpd_tls_session_cache_timeout = 3600s
smtpd_use_tls = yes
virtual_gid_maps = static:5000
virtual_mailbox_base = /var/mail/vhosts
virtual_mailbox_domains = example.com
virtual_mailbox_maps = hash:/etc/postfix/vmailbox
virtual_minimum_uid = 100
virtual_uid_maps = static:5000
postconf: warning: /etc/postfix/main.cf: unused parameter: smtpd_sasl_auth_clients=yes
```

Figure 39.

- Check Dovecot configuration:

```
doveconf -n
```

```
root@mail:~# dovecot -n
# 2.3.21 (47349e2482): /etc/dovecot/dovecot.conf
# Pigeonhole version 0.5.21 (f6cd4b8e)
# OS: Linux 6.8.0-88-generic x86_64 Ubuntu 24.04.3 LTS ext4
# Hostname: mail.example.com
auth_mechanisms = plain login
mail_location = maildir:/var/mail/vhosts/%d/%n
mail_privileged_group = mail
namespace inbox {
  inbox = yes
  location =
  mailbox Drafts {
    special_use = \Drafts
  }
  mailbox Junk {
    special_use = \Junk
  }
  mailbox Sent {
    special_use = \Sent
  }
  mailbox "Sent Messages" {
    special_use = \Sent
  }
  mailbox Trash {
    special_use = \Trash
  }
  prefix =
}
passdb {
  args = scheme=PLAIN username_format=%u /etc/dovecot/users
  driver = passwd-file
}
protocols = imap pop3 lmtp imap lmtp pop3
service auth {
  unix_listener /var/spool/postfix/private/auth {
    group = postfix
    mode = 0660
    user = postfix
  }
}
ssl = required
ssl_cert = </etc/ssl/mail/mail.example.com.crt
ssl_client_ca_dir = /etc/ssl/certs
ssl_dh = # hidden, use -P to show it
ssl_key = # hidden, use -P to show it
ssl_prefer_server_ciphers = yes
userdb {
  args = uid=vmail gid=vmail home=/var/mail/vhosts/%d/%n
  driver = passwd-file
}
```

Figure 40.

- ```
systemctl status postfix
systemctl status dovecot
systemctl status opendkim
```

```
Dec 19 21:42:34 mail.example.com dovecot[1228]: auth: passwd-file:user@example.com::!L:~NwvSpRjKtA8AAAAAAAAAAAAAAAAAB: user not found from userdb
Dec 19 21:42:34 mail.example.com dovecot[1228]: jmap[5166]: Error: auth-master: login: request [922677953]: Login auth request failed: Authenticated user not found from userdb, user lookup 3
Dec 19 21:42:34 mail.example.com dovecot[1228]: imap-login: Disconnected: Internal login failure (pid=5158 id=1): user=user@example.com, method=PLAIN, rls=::1, lip=::1, mpid=5166, TLS, se=
Dec 19 21:42:36 mail.example.com dovecot[1228]: imap-login: Disconnected: Connection closed (no auth attempts in 25 secs): user=, rpi=::1, lip=::1, TLS: Connection closed: session, ca=Cw659
Dec 19 21:45:25 mail.example.com dovecot[1228]: auth: passwd-file:user@example.com::!L:~NwvSpRjKtA8AAAAAAAAAAAAAAAAAB: user not found from userdb
Dec 19 21:45:25 mail.example.com dovecot[1228]: jmap[5208]: Error: auth-master: login: request [922677953]: Login auth request failed: Authenticated user not found from userdb, user lookup 3
Dec 19 21:45:25 mail.example.com dovecot[1228]: imap-login: Disconnected: Internal login failure (pid=5208 id=1): user=user@example.com, method=PLAIN, rls=::1, lip=::1, mpid=5208, TLS, se=
Dec 19 21:48:33 mail.example.com dovecot[1228]: imap-login: Disconnected: Connection closed: SSL_accept() failed: error:BA080418:SSL routines:tslib alert unknown ca: SSL alert number 48 (no
```

- Check listening ports:

```
ss -tlnp | grep -E ':(25|587|465|143|993|110|995)'
```

```
MAIL FROM:user1@example.com
250 2.1.0 OK
RCPT TO:user2@example.com
250 2.1.5 OK
DATA
354 End data with <CR><LF>.<CR><LF>
Subject: Orange tree
This is dungeon with apples.
.
250 2.0.0 Ok: queued as 6685C141129
QUIT
221 2.0.0 Bye
Connection closed by foreign host.
root@mail:~# echo "TEst" | mail -s "TEst subject" user2@example.com
root@mail:~# sudo tail -f /var/log/mail.log
2025-12-10T21:31:33.964124+00:00 mail postfix/qmgr[4913]: 6685C141129: removed
2025-12-10T21:31:35.667872+00:00 mail postfix/smtpd[4921]: disconnect from localhost[127.0.0.1] ehlo=1 mail=1 rcpt=1 data=1 quit=1 commands=
2025-12-10T21:31:51.217440+00:00 mail postfix/pickup[4912]: 34DE81422FC: uid=0 from=<root@mail.example.com>
2025-12-10T21:31:51.221105+00:00 mail postfix/cleanup[4930]: 34DE81422FC: message-id=<20251210213151.34DE81422FC@mail.example.com>
2025-12-10T21:31:51.221644+00:00 mail opendkim[1229]: 34DE81422FC: no signing domain match for 'mail.example.com'
2025-12-10T21:31:51.221689+00:00 mail opendkim[1229]: 34DE81422FC: no signing subdomain match for 'mail.example.com'
2025-12-10T21:31:51.222309+00:00 mail opendkim[1229]: 34DE81422FC: no signature data
2025-12-10T21:31:51.236755+00:00 mail postfix/qmgr[4913]: 34DE81422FC: from=<root@mail.example.com>, size=375, rcpt=1 (queue active)
2025-12-10T21:31:51.245253+00:00 mail postfix/virtual[4939]: 34DE81422FC: to=<user2@example.com>, relay=virtual, delay=0.09, delays=0.08/0/0/0.01, dsn=2.0.0, status=sent (delivered to maildir)
2025-12-10T21:31:51.245331+00:00 mail postfix/qmgr[4913]: 34DE81422FC: removed
```

- To copy all config files I used `cp` and shared folder:



```
root@mail:~/share# cp /etc/hosts \
/etc/postfix/main.cf \
/etc/postfix/master.cf \
/etc/postfix/vmailbox \
/etc/dovecot/dovecot.conf \
/etc/dovecot/users \
/etc/dovecot/conf.d/10-mail.conf \
/etc/dovecot/conf.d/10-auth.conf \
/etc/dovecot/conf.d/auth-passwdfile.conf.ext \
/etc/dovecot/conf.d/10-master.conf \
/etc/dovecot/conf.d/10-ssl.conf \
/etc/openssl.conf \
/etc/openssl/KeyTable \
/etc/openssl/SigningTable \
-t ~/share/config_files
```

Figure 43. Copying configs to shared folder.

## Security takeaways

### TLS/SSL Encryption

TLS/SSL was enabled for SMTP submission, IMAP, and POP3, ensuring all communication between clients and the server is encrypted. This prevents credential theft and protects message confidentiality during transit.

### Authentication Mechanisms (SPF, DKIM, DMARC)

SPF records were configured to specify who is allowed to send to the host.

DKIM was implemented through OpenDKIM to cryptographically sign outgoing mail, and DMARC policies enforced connection between SPF and DKIM.

These tools collectively mitigate spoofing & improve trust with other receiving servers.

### Port Security and Firewall Configuration

The firewall was restricted to essential ports: 25 (SMTP), 587 (submission), 993 (IMAP-SSL), and 443 (HTTPS for Roundcube).

All other ports were closed to reduce attack surface and prevent unauthorized access.

## Discussion

How Email Security Protocols Work Together?

SPF restricts unauthorized senders, DKIM signs messages to verify integrity, and DMARC ties both together, defining how receiving servers should handle failures. Combined with TLS, these protocols provide layered protection against spoofing, interception, and identity fraud.

### Real-World Applications and Use Cases

This architecture mirrors what many organizations deploy: custom mail servers for internal communication, secure messaging environments, and hybrid setups that combine local mail handling with cloud spam filtering.

## Summary

The lab demonstrated how to set up a lab environment the closest to a real-world implementation of local email servers. During the lab, I configured main interfaces and tools to create the environment for secure email messaging, starting from DNS server configuration to configuring a web-app for managing end-users' mails.

## References

- Postfix Documentation: <http://www.postfix.org/documentation.html>
- Dovecot Wiki: <https://doc.dovecot.org/>
- OpenDKIM Documentation: <http://www.opendkim.org/>
- Roundcube Documentation: <https://github.com/roundcube/roundcubemail/wiki>